CIMA Strategic Case Study

Mock Exam 4

Rotomyne

Suggested Solutions

How to use Astranti Solutions

What an Astranti Solution is for

Our solutions are designed to be **perfect answers**. That means that they cover every possible point in depth in order to serve as a learning tool.

This is so that when you review your own solutions against ours, you can see what you got right and what you got wrong. What you covered in enough depth and what you missed out.

What an Astranti Solution is NOT for

Our solutions are not an example of what you should be able to achieve in the exam!

When you sit the real exam, you simply **won't have the time to write an answer as long as ours**. Perhaps you'll cover 75% of it and this is completely fine as long as you have answered the question.

CIMA solutions

You may have noticed our solutions differ from the CIMA solutions. CIMA solutions to past exams are simply examples of a script that might have passed.

They do not prepare, guide or teach you how to improve your answers and increase your chances of passing. Ours do.

Industry examples

You'll notice that we've included real world industry examples throughout the solutions. This is something you should be doing in your own solutions.

Examiners repeatedly emphasise the **importance of wider industry knowledge** and industry examples are the best way to earn the available marks in this area.

Task 1 - Solution

To: Martin Jacobs, CFO
From: Senior Manager (SM)
Subject: Briefing note - Risk register, growth strategies and data security

Hi Martin,

Thank you for your email. Following your request, please find below my discussion on the risk register and the methods used to identify risks. Please also find my discussion of the different growth strategies available to Rotomyne. Lastly, I have included a discussion of the controls that should be in place to improve data security.

Sub-task (a) - 55%: Risk register and risk treatment

1ai - Risk register, principal risks log reuse & new risks identification

Risk register contents

Rotomyne's recent risk audit has highlighted several gaps in the company's risk management process, including the need for a formal risk register. A risk register would allow Rotomyne to systematically document and track risks associated with its global lithium mining operations. This tool typically follows a structured format, allowing for the identification, measurement, treatment, and monitoring of risks in a centralised document.

Logging the risk and assigning ownership

The first step in creating the risk register is to log the risk and assign ownership. For Rotomyne, risks such as exchange rate fluctuations and environmental regulations need to be clearly recorded.

The owner of the risk is commonly a manager or director. For instance, responsibility over exchange rate risks might be held by the CFO or by a senior finance manager.

The risk title summarises the nature of each risk, while the risk description details the causes and potential impact on the business. For example, "exchange rate risk" could describe how fluctuations in currency values affect revenues from international sales. Each risk should also be dated when logged to track its development.

Assessing and measuring the risk

After logging, each risk should be assessed and measured. This involves assigning a likelihood score to indicate the probability of the risk occurring and an impact score to quantify the damage it could cause to Rotomyne. These scores are commonly on a scale of 1 (low impact/likelihood) to 5 (extremely impactful/likely).

For example, environmental regulations could carry a high impact score if stricter rules lead to delays or shutdowns in mining operations. Multiplying likelihood and impact gives a risk score that helps in prioritising which risks require immediate attention.

Treating the risk and setting targets

Once measured, each risk must be treated and managed, and the date that mitigations are implemented should be recorded.

Rotomyne already takes several mitigation actions, such as using hedging strategies to reduce currency risks and complying with environmental standards. These actions help lower the residual risk score, which is based on measurements of the risk after mitigating action has been taken. For instance, the treasury team's hedging efforts reduce the likelihood of financial damage from exchange rate volatility.

Rotomyne should also set a target risk level, which is an acceptable level of risk aligned with the company's strategy.

If the residual risk remains above the target, further action must be taken. These further mitigation actions might include controls or an action based on one of the TARA (Transfer, Accept, Reduce, Avoid) risk management strategies mentioned in Iresh Jayawardena's email.

Monitoring and reviewing risks

A critical part of the process is scheduling the monitoring and reviewing of risks, either on a regular basis (e.g. weekly, monthly, quarterly, annually). These reviews should also be noted and dated in the risk register when they have occurred, so that the ongoing monitoring is also recorded and to ensure that Rotomyne continuously assesses its risk exposure, especially for evolving threats like cybersecurity risks. Frequent reviews help mitigate emerging risks before they escalate.

How the principal risk log fits into the risk register

Rotomyne's principal risks log has some of the key sections recommended in a risk register. These are the 'Risk title' and 'Risk description' (as can be derived from the 'risk impact' column) as well as the 'Mitigation actions' that help reduce the risk (as seen in the 'risk mitigation' column). However, the principal risk table notably lacks formal risk scoring and explicit ownership assignment.

Once the risk information from our principal risks log has been transferred to the risk register then the ratings for the 'Risk likelihood' and 'Risk impact' can be assessed (albeit qualitatively). These ratings then help calculate an overall rating i.e. the 'Risk score'.

The risks that currently appear in our Principal Risks log that would score highly in either likelihood or impact (and possibly both) are exchange rate fluctuations, which would affect our costs and revenues, and the risk relating to environmental regulations. Once the 'Mitigation actions' have been taken into consideration, then we would be in a position to determine the 'Residual risk score'. This would also help in prioritising the risks.

Risk ownership is key as it would assure accountability for the risks. Risk ownership is implied in our principal risks log via the 'risk mitigation' column. For example, the treasury team is allocated to manage exchange rate fluctuations and the production team is responsible for the mitigation of price risk.

We should clearly assign a 'Risk owner' to the risks which should ideally be at the senior management or director level e.g. the Production Director (Andrey) should be the risk owner for the environmental regulation risk and the CFO (Martin) the risk owner for the risk of exchange rate fluctuations. Also, if we had an IT director he or she could be the risk owner for the cyber risk that was identified by the Risk Audit.

1aii - Risk identification methods

Identification of new risks for the risk register

Although they do not appear on our principal risks log, missing key risks have been pointed out by the Risk Audit. For example, cyber risk is a key risk that we have not identified.

The risk register would help identify such new and important risks via its implementation and usage. In order to adequately identify (and mitigate) all key risks, the risk register should be a comprehensive risk log, The following are some common methods we could use to identify risks:

PESTEL and SWOT frameworks

PESTEL (Political, Economic, Social, Technological, Environmental and Legal) and SWOT (Strengths, Weaknesses, Opportunities, Threats) analyses provide useful headings to help identify and think about the risks that may face us.

For instance, the cyber risk could have been identified whilst applying the 'technological' aspect of the PESTEL framework to the risk register.

Interviews and questionnaires

Risk identification can also be driven by gathering insights from key department heads. For example, the production team might identify operational risks related to equipment maintenance or labour shortages in mining operations. Regular interviews and questionnaires with managers across departments would help highlight risks specific to each area of the business.

This could also be supported by input from our Risk Committee. It is also common for the risk committee to be responsible for the control of the whole risk identification process.

Internal and external audits

The recent risk audit that flagged missing risks (such as cybersecurity threats) demonstrates the value of regular internal audits. These audits can identify emerging risks in Rotomyne's core areas, such as production and treasury.

Our external auditors could provide additional expertise, particularly in identifying risks Rotomyne may not be fully equipped to assess, like cybersecurity vulnerabilities.

External advisors

We could also consider employing external advisors, such as risk consultants, to identify risks. They may have broader knowledge from different industries and sectors that we are not familiar with. They could also advise on how to mitigate the risks they identify. This option may be expensive, but may be beneficial as it would help us to become more risk aware.

Brainstorming

Rotomyne's Board and senior management could benefit from brainstorming sessions to explore new risks in a more informal setting. These sessions could focus on the risks associated with new markets or products, such as the increasing demand for electric vehicle batteries. Brainstorming would enable a variety of viewpoints to be considered, making it easier to anticipate potential risks before they become significant issues.

Conclusion

Implementing a risk register would significantly enhance Rotomyne's ability to manage its risks in a structured and transparent way. By incorporating missing risks like cybersecurity threats, assigning clear risk ownership, and applying formal risk scores, Rotomyne can better prioritise its risks.

This tool will enable Rotomyne to stay agile in a competitive market, ensuring that the company is well-prepared to manage the risks associated with its global growth, particularly in sectors such as electric vehicle battery production.

Sub-task (b) - 25%: Growth strategies

1bi - Growth strategies

Growth strategies

Rotomyne is keen to grow to keep up with global demand for its current products, but it could also pursue other product or market growth options. However, as has been mentioned, the various growth options have varying levels of risk that Rotomyne would need to be prepared to accept.

Ansoff's Matrix could help us to identify different approaches for possible strategic growth and their risk level. Ansoff identified four growth strategies that are derived from the mix of existing or new products and markets (customers).

Market penetration (existing market, existing product)

A market penetration strategy would involve seeking growth with our existing products in our current market segments, aiming to increase Rotomyne's market share. Typical examples of market penetration strategies include reducing prices, increasing advertising or further differentiating the product. This is also usually the least risky strategy since it builds upon many of the company's existing resources and capabilities.

In our case, as demand for electric vehicles is increasing and since most transactions are based on a spot price negotiated and agreed at the time of sale, we could focus on market penetration and try offering bulk purchase discounts on orders of our battery grade lithium hydroxide. The aim of the market penetration strategy would be to win market share from our direct competitors (e.g. Lithdig). However, because lithium is a commodity, it is difficult for us to differentiate ourselves from our competitors, and pricing strategies can be replicated. As such, this strategy might not be entirely effective.

We could also offer exceptional customer and technical support to our existing customers to encourage them to purchase more. For example, Albemarle (a leading US-based lithium producer) has been working closely with the US electric vehicle manufacturer Tesla to supply it with high quality lithium hydroxide for its battery production. This has allowed Albemarle to grow through market penetration.

Market development (new market, existing product)

A market development strategy seeks growth by selling existing products to new customers. These customers could be found in new market segments (i.e. segmented by demographics or distribution channel). They could also be found in new geographical markets (and reached by an overseas expansion).

The risk of these kinds of strategies is usually considered to be medium as market development involves exploring a new market that is somewhat unknown to the organisation but the organisation will continue to sell a product that it already knows.

In our case, we could enter new geographic markets where electric vehicle production is growing, such as Asia and Europe. We have customers in over 90 countries, so we would have to research to find which countries to expand into that we have not already. Considering our broad range of existing markets, geographical expansion may not yield significant new growth at this stage unless a new market emerges, such as a new large scale production plant in a previously less desirable geographical market.

Another market that Rotomyne could explore might be the renewable energy industry. In order to ensure continuous energy availability, wind and solar energy production facilities need energy storage capabilities. This growing market could be a new avenue to sell our lithium products at large scales. Furthermore, this strategy would comply with our mission, helping to build 'a world that is clean, healthy and sustainable.'

Product development (existing market, new product)

A product development strategy aims to develop new products targeted at existing markets (or market segments). This enables sales to existing loyal customers and capitalises on the company's brand strengths. The risk of this strategy is medium as it is an existing market, but a new product that has the potential to fail, wasting the investment and the opportunity.

In our case, we could further fund research into ways to improve the quality of lithium we produce. For example, we could research ways to enhance the purity and performance characteristics of our lithium carbonate and lithium hydroxide, which could help meet the evolving needs of battery technology.

As these enhanced products are essentially just a variant of our current product, a product development strategy is less risky from an investment and customer adoption perspective.

Diversification (new market, new product)

A diversification strategy seeks growth by developing new products for new markets. There are two types of diversification, related and unrelated. Related (or vertical) diversification is where the new product and market is related to existing operations. Where the new product and market is unrelated it is known as unrelated (or horizontal) diversification.

A diversification strategy is very risky due to the lack of experience and knowledge of both the product and market area. This risk tends to increase if the diversification is unrelated to the products and markets about which the organisation has knowledge and experience.

In our case, we could pursue vertical diversification by moving downstream. For example, by manufacturing or recycling batteries to capture more value from our supply chain.

A relatively low-risk product that Rotomyne could explore would be repurposing some of the waste materials from our refinery processes as industrial materials. For example, some of the waste generated during the process might be usable as an additive in cement, which could be sold to the construction industry. Not only would this byproduct provide a small revenue stream, but it would also provide a method for disposing of some otherwise potentially hazardous waste, which would also comply with our mission regarding sustainability.

We could also diversify into other renewable energy materials. For example, sodium for wind turbine sodium-ion batteries. These options would require a lot of capital investment, but they would also address the issue of finite available lithium, which may become more scarce in the future as sources worldwide begin to dwindle and risk the future of Rotomyne.

For the same reason, it may be worth considering opportunities for unrelated diversification, but with the commensurately increased risk associated with that growth strategy.

Sub-task (c) - 20%: Data security controls

1ci - Data security controls

Controls to improve data security

We live in the digital era where data could be accessed remotely and stolen or corrupted. The Risk Audit's concerns with regards to data security are well founded. For example, in 2023, data breaches compromised the personal information of millions of people around the world. Some of the biggest UK victims in 2023 included Capita, Arnold Clarke, the PSNI/UK Police, the University of Manchester and the NHS (National Health Service), demonstrating that larger companies and institutions are regularly targeted by malicious actors and despite having defences in place can still be breached.

Below are a number of steps that Rotomyne would need to take to better protect the data in our care:

Access

A key control to improve data security involves limiting employee access. This should not prevent our employees from doing their job, but should instead ensure employee's access is limited to the information they need in order to fulfil their daily roles. No additional access to data should be given.

For example, not every person in the company would need access to information about the purchase history of our lithium products. This information should be limited to a select few people (on our finance and sales teams, for instance) who could learn from the insights this data provides. These individuals could then share their broader insights with other individuals within Rotomyne.

Some sensitive data that is used semi-regularly should also be further secured, such as customer financial information or our own confidential information. This might be set up with Multi-Factor Authentication (MFA), where senior employees or managers are alerted or need to give specific permission for employees to access certain files and areas.

Segregation of duties

In line with limiting access, Rotomyne would also need to ensure that no one individual has too much access to systems or data. We could achieve this via the segregation of duties. The more people have to collaborate to gain access to meaningful data (such as customer orders) or customer sensitive data (such as address, payment details), the less likely a data breach is to occur.

Awareness of sensitive data

It is vital for Rotomyne that our employees are aware of what constitutes sensitive data (e.g. ethnicity, religion, biometric data, health data). In terms of the information that we hold, in reality only a small proportion of it would be classed as sensitive, which we would therefore need to have strict controls over. Identifying sensitive data would help us maintain cost effective controls and prevent unnecessary measures being put in place to protect freely available data such as our website hits or social media followers.

Training

In line with the awareness of sensitive data, Rotomyne should train staff to ensure everyone is aware of what sensitive data is (e.g. as per GDPR), the risk that this data poses (e.g. potential fraud) and how it should be controlled. Ensuring staff adopt good IT habits like developing strong passwords, securing hardware and information overnight, and never leaving computers unattended would also mitigate the threat of risks here.

Threats from phishing and social engineering remain a significant vulnerability for many organisations that have otherwise excellent controls and cybersecurity defences. Our staff, especially those with more access to our systems, must be trained to identify attempts to manipulate them through tactics like phishing calls or emails.

Up-to-date systems

Another key factor to prevent any data security issues would be to maintain up-to-date systems, IT is constantly evolving and hackers are finding new techniques to attack systems and either steal information or corrupt data. This would cause major disruption.

As a result, software developers are frequently upgrading the systems they support to make sure any potential flaws are resolved before they become an issue. This is only done for current operating software, so it would be important to ensure that we use up-to-date software, providing us with more protection from our software providers.

Antivirus and anti-malware software that automatically updates, should also be used on all computers on the company's network. This may be a challenge as our mining operations are located in six countries, but this is necessary.

Back-ups

As part of good business continuity planning, regular data back-ups should be made. These back-ups should be kept at a secure external location. This way, if a malicious actor or malware breaches our system and either destroys or otherwise limits our access to our systems, we will be able to restore our systems with less disruption. As such, it is also important that these back-ups are regularly updated so that it minimises the data lost by our system going down.

I hope this helps, but please do get in touch if you have any further questions.

Thanks,
SM

Task 2 - Solution

From: Senior Manager (SM)
To: Martin Jacobs, CFO
Subject: Project risk and currency risk

Hi Martin,

Thank you for your email. In response to your request, I will discuss the benefits of a Post Completion Audit (PCA) and recommend controls to manage a project's time, cost and quality. Additionally, I will discuss the different types of currency risks as well as the methods to mitigate the foreign exchange transaction risk.

Sub-task (a) - 55%: PCA and project control

2ai - Post Completion Audit

A Post Completion Audit (PCA) (also known as a Post-Implementation Project Review/Audit) is performed at the end of a project and involves a review of the project against its original objectives to learn from the project for the future.

From a business perspective it might include identifying lessons about the market, product or project management approach taken. From a financial perspective it can be useful to review the project against initial cost estimates and assumptions so that lessons can be learnt when budgeting for similar projects in the future.

A project's PCA provides the mechanism whereby experience of past projects can be fed into the entity's decision-making processes as an aid to the improvement of future projects. In particular, it has been suggested that regularly performing PCAs at the end of major projects could help Rotomyne to better manage its risks, particularly those concerning the issues that arose in the most recent project.

Benefits of a PCA

There are a number of potential benefits of regularly performing PCAs, such as those discussed below:

1) Quality decisions

Regularly performing PCAs will improve the quality of decision-making by providing a mechanism whereby past experience can be made readily available to decision-makers involved in current or future projects.

Using experience of past similar situations to help guide decisions greatly reduces the chance of making the wrong choice, which could otherwise prove costly or would require re-work.

Caroline Nguyen Ngoc specifically referred to the issues experienced in the recent IT system and warehouse relocation projects. When the next IT project is planned and executed, our decision-makers will be able to review the PCA for this recent project, including any accounts by the decision-makers in that project, and they will then be better equipped to avoid those same mistaken decisions.

2) Realistic forecasts

It encourages greater realism in project appraisal by providing a mechanism whereby past inaccuracies in forecasts are made known. This would help to ensure that deadlines are realistic and achievable. Setting forecasts that would never be reached can act as a de-motivating factor or could lead to failure of a project or delays especially when there are dependent tasks.

Where the same project team is involved in multiple projects, this review process will help them to reflect on the assumptions that fed into the original forecasts and then to generate more realistic ones in the future. For instance, with the recent issues with the warehouse relocation and the IT project, these exceeded both the projected costs and timeframes, and this is part of a larger pattern of cost overruns and project delays. As such, there is likely an issue of inaccurate forecasts, flawed project execution or both.

3) Better project control

Routinely undertaking PCAs will provide a means of improving control mechanisms by formally highlighting areas where weaknesses have caused problems. In order to enable the PCAs to be conducted after each project, project managers might be required to maintain a thorough project plan, risk log and quality log during each new project, which can then be used during the review, rather than relying on other documentary evidence or on individual personal accounts of the project's issues.

Using this evidence, along with other sources, to identify and then address weaknesses in the project controls is key if we are to make a success of future projects at Rotomyne.

4) Quicker project modifications

Conducting PCAs enables speedy modification of future underperforming or overperforming projects by teaching us how to identify the reasons for deviation from the expected performance. The elements that lead to underperformance can then be corrected as soon as they are spotted, or those indicating overperformance can be nurtured and replicated.

Once these PCAs are regularly occurring, we can use the results from our attempts to modify and correct past deviating projects in order to inform our management of current and future projects. Through this process, we will be able to develop tried and tested methods for correcting projects.

5) Identification of project success and failure factors

By reviewing past projects and their failures, it will become easier to identify the warning signs that a project is doomed to failure. And, in these cases, we will be better prepared to terminate the project, limiting the resources we sink into the project past this point. Knowing when to exit helps focus attention on more profitable ventures.

Similarly, regular project reviews will highlight reasons for successful projects that may be important in achieving greater benefits from future projects. Having a recipe for success by analysing successful projects reduces the risk of failure. At times, success may be to do with a combination of factors that may not be apparent without the benefit of hindsight from other projects.

2aii - Recommended project controls for time, cost and quality

Projects are constrained, and therefore assessed, on time (i.e. implemented without slippage), cost (i.e. within budget), quality (i.e. faults within expected parameters) and scope (i.e. the project has achieved its overall and specific objectives). Project controls need to be put in place in order to ensure that these constraints are adhered to and any deviations are minimised or avoided entirely.

Evidently there have been flaws in the controls systems in place for past projects considering the consistent record of poor project implementation in recent years. The controls should be reassessed and supplemented, and as such, the following suggestions may assist with this revision and help to address issues with adhering to the project constraints.

Project time (schedule) controls

Project plan

It is noted that Rotomyne's projects usually have to be delivered to tight deadlines.By tight, this typically indicates a lack of slack or buffer time, so a single small delay can have a knock-effect that delays the entire project, rather than being absorbed with other tasks being completed more quickly to compensate for earlier delays.

One essential control to implement in order to meet tight deadlines is ensuring that there is a robust project plan that the project manager is consulting frequently. Using project planning tools such as a Gantt chart will help to identify where there may be bottlenecks or overlapping tasks that could impact each other.

The progress of the project should also be regularly checked against anticipated project milestones, as this will help to identify when the project has started to fall behind and the ultimate impact this will have on the completion date, so that other key project stakeholders can be kept informed about its progress.

The current project plan should be updated when these delays are identified.

External party controls

Often on a project, there may be external parties involved in helping us to fulfil our projects. For instance, we may have hired an external IT company to help us with the launch of the new IT system or with the preparation of the new warehouse for the planned relocation.

When negotiating contracts with these external organisations, controls should be included in the contract that allow us to carefully monitor their fulfilment of the terms of the contract. And, if we feel that they are needed, we might even need to include some provisions for failing to uphold their ends of the contracts, such as financial penalties for delayed delivery of services or products, or strict limits of the amount we are willing to pay, to limit our exposure to cost increases leading to cost overruns.

Cost controls

Budgeting and variance analysis

While our projects will already have budgets, evidently these are not being adhered to strictly enough or the budgets themselves were unrealistic, given the regular cost overruns in recent years leading to both the IT project and warehouse relocation costing twice what was budgeted. In terms of controls, these should focus on monitoring and enforcing the existing budget, though the approach to generating the budget may be updated if that is the source of the issue.

Based on the budgets that are generated, we can perform variance analyses on a regular basis (e.g. monthly) to identify deviations. Significant variances would need to be reported and analysed. If necessary, authorised corrective action should then be taken (e.g. more money allocated to a task). Following this, the revised budget should be calculated. The budget control process would then be repeated.

Spending authorisation controls

Following on from the point above, a spending authorisation system would ensure that only approved expenses are made during the project. This control would prevent unapproved or unnecessary spending, which can cause projects to exceed their budgets. Only designated personnel, such as the project manager or finance officer, should have the authority to approve expenditures. Purchase orders should also be reviewed and approved in accordance with the pre-set budget allocations.

While this approach may introduce restrictions and delays to the projects, while the budget overruns are so severe, a strict system in place like this may be necessary.

Change control for budget adjustments

Planned projects do change when exposed to the real world, so adjustments sometimes will have to be made to the projected budgets. When project scope changes or unforeseen circumstances arise, a formal process for adjusting the budget will ensure that any additional spending is properly reviewed and approved and will help to prevent uncontrolled budget increases.

Quality controls

Project quality plan

Our future project would need to be delivered to the required quality (i.e. meeting our original objectives). This would mean that all the business requirements and specifications for any future project would have to be met (e.g. highly trained staff and availability of quality IT infrastructure).

Poor quality would cause issues for us, especially as we would be competing for customers that could go to our competitors if they are not happy. Previously, we have had quality issues on our projects and we may have lost customers on the back of this as well as suffered reputational damage.

We can control project quality through the use of a project quality plan (PQP). This document would detail the standards that are required for the project and how these standards would be tested. Periodic quality reviews should be done (via testing). If the required quality is not met (within a certain limit), then corrective action would need to be done (e.g. re-work).

Scope controls

Change control processes

Scope creep can lead to other issues, such as problems abiding by the budget, hitting the deadline or maintaining the quality standard that was originally agreed. As such, this should ideally be avoided unless absolutely necessary.

To reduce the amount of scope change once the project has started, implementing controls to monitor and limit the number and scale of changes should be enacted. For instance, where the project manager or other stakeholders wish to make changes to the scope of the projects, or to other elements, they must gain approval from the project's sponsor or another suitable senior project stakeholder.

Sub-task (b) - 45%: Currency risk & currency risk management

2bi) Foreign currency risk types

Currency risk

We have exposure to many currencies due to our mines and customers being located globally. Our profits and foreign currency reserve are being negatively affected by fluctuations in the different currencies with which we do business. There is also additional currency risk due to the exchange rate volatility from the politically unstable countries where two of our mines are based. As such, we need to better understand currency risk and seek to mitigate it, in particular the foreign exchange transaction risk.

Currency risk is the risk of potential exchange rate movements impacting a business. This risk could result in either an exchange rate gain (upside risk) or loss (downside risk). Adverse exchange rate movements are more of a concern as this would negatively affect the value of our company's assets, investments and their related interest and dividend payment streams, especially those securities denominated in foreign currency. There are three main types of currency risk: economic risk, translation risk and transaction risk.

Economic risk

Economic currency risk refers to the long-term impact of exchange rate fluctuations on a company's market value or competitiveness, as changes in currency rates can affect future cash flows, costs, and overall financial stability. Economic factors include economic growth, interest rates, exchange rates and the inflation rate.

We are exposed to economic risk due to having our mines in 6 countries and customers in 90 countries. Rotomyne faces economic currency risk if sustained changes in exchange rates make its lithium products more expensive for foreign customers, potentially reducing demand. For instance, if the P$ strengthens significantly, Rotomyne's lithium products could become more expensive for customers paying in weaker foreign currencies, which could hurt its competitive position in the global market, especially in price-sensitive sectors like electric vehicle manufacturing.

Translation risk

Translation risk is an accounting concept relating to the value of assets or liabilities held in foreign currencies. Changes in the exchange rate over time will alter the value of assets or liabilities when converted back into the business' home currency when reporting in the financial statements.

Although there could be currency gains (upside risk), the downside risk is that the whole balance sheet value could deteriorate in the home currency despite the overseas operations actually being successful. We are exposed to translation risk as our consolidated financial statements' reporting currency (home currency) is P$.

Transaction risk

Transaction risk is the risk that an exchange rate will change unfavourably over time. It is associated with the time delay between entering into a contract and settling it and then having to pay more than we were expecting because rates have moved unfavourably. Transaction risk arises when a company engages in foreign currency transactions, such as paying suppliers or receiving payments from customers in different currencies, and fluctuations in exchange rates affect the value of these transactions.

The greater the time differential between the entrance and settlement of the contract, the greater the transaction risk, because there is more time for the two exchange rates to fluctuate. This risk impacts actual cash-flows and reduces profitability.

We are exposed to transaction risk as we trade internationally, have credit sales and a trade receivables balance. For example, if Rotomyne exports lithium products to a country where the currency depreciates against the Porrland dollar (P$), the company could receive less revenue in P$ than anticipated, impacting profitability.

2bii) Foreign exchange transaction risk mitigation

There are a number of methods that we could employ to manage transaction risk, some of which are detailed below.

Invoice in home currency

We could inform our suppliers and customers that our invoicing is to be done in our home currency (P$) going forwards. Two of our mines are located in countries where the currency is quite volatile, as such we should prioritise those countries.

This is an internal hedging method and it is the simplest way of dealing with transaction risk. By doing this, we would transfer the transaction risk onto our suppliers and customers. However, unless we have strong bargaining power over our suppliers and customers, we might not be able to demand this.

Money market hedge

A money market hedge (MMH) avoids future exchange rate uncertainty by making the exchange at today's spot rate instead. This is done by depositing/borrowing the foreign currency until the actual commercial transaction cash-flows occur. There are two types of money market hedge - foreign currency payment and foreign currency receipt.

In the case of a future foreign currency payment (e.g. to our suppliers), our home currency (P$) would be converted immediately at the rate today (i.e. spot rate), and monies invested in a foreign currency bank account, until required for payment in the future. For a foreign currency receipt (e.g. from our customers), a loan would be taken out in a foreign currency, and would then be converted at the spot rate. When the monies are received in the foreign currency they are used to pay off the foreign currency loan.

As such, for both payment and receipts, the exchange rate used would be the spot rate, and hence the risk of rates changing by the payment or receipt date would be removed. As can be seen, it is rather complex to use a MMH to reduce transaction risk. However, if we have the expertise internally to set up such a hedge or pay a third party (e.g. a bank) to do this, then it could be an effective way of eliminating the transaction risk.

Forward contract

If a supplier invoices us in a foreign currency, then we could ask our bank to set up a forward contract relating to the invoice. This would allow us to buy the foreign currency (from our bank) on a fixed future date (i.e. the date we need to make payment) at a predetermined rate, i.e. the forward rate of exchange. Since our bank would effectively guarantee us a specific exchange rate on the transaction (payment settlement) date, we would be protected from the transaction risk even if exchange rates fluctuated.

In the case of a customer sending us a payment of a foreign currency in the future, we would set up a forward contract to sell this amount at the forward date. This would lock in the exchange rate for the conversion back into P$. Forward contracts are simple to arrange, flexible and are a relatively cheap way of mitigating transaction risk.

Futures contract

A futures contract allows us to buy or sell the foreign currency on a fixed future date at a predetermined rate. It is similar in principle to a forward contract. However, it differs from a forward contract in that the futures contract is for a standardised amount, which is traded on currency exchanges (e.g. LIFFE, CME, MICEX). So, if the foreign currency is available as a currency for a futures contract, then we could use a futures contract to hedge transaction risk.

A futures contract is more complex to arrange. Also, because it is for a standardised amount, it is more appropriate for the hedging of large currency amounts. In addition, the currency amount covered by a futures contract might not hedge our transaction risk completely. That said, it still provides a method to reduce or eliminate our transaction risk.

Our competitors use futures contracts to hedge transaction risk. For example, Ganfeng Lithium Group (a large Chinese lithium producer) uses futures contracts on the Guangzhou Futures Exchange to hedge against price volatility in lithium carbonate. This approach helps them to stabilise their revenue and manage the financial impact of currency movements.

Currency options

The above methods should allow us to eliminate our transaction risk. However, if the exchange rate moves in our favour (i.e. upside risk) we would not be able to benefit from this favourable movement. This is where currency options are more useful. They allow the use of a predetermined exchange rate without being locked into it.

As such, currency options are a flexible hedging method as they cover the downside risk, but allow the benefit of a favourable movement (of the P$ against the foreign currency). In this way, we could for example reduce our overall purchase costs for supplies. The disadvantage is that a currency option has an expensive premium that has to be paid even if the currency option is not exercised (i.e. lapses).

I hope my responses have sufficiently addressed all your queries.

Kind regards,

SM

Task 3 - Solution

To: Martin Jacobs, CFO
From: Senior Manager (SM)
Subject: Briefing note - International expansion options and business valuation

Hi Martin,

Thank you for your email. As requested, I have evaluated the different options for expansion into Bigland and recommended the most suitable option. I have also discussed the valuation methods used and their suitability as well as recommending an approximate valuation.

Sub-task (a) - 55%: International expansion

3ai - International expansion: Options analysis

Introduction

We are looking at pursuing growth through market development, selling battery-grade lithium hydroxide to new international customers in Bigland. As suggested, there are multiple options/methods we could use to expand/grow internationally, such as the following.

Option 1 - Organic growth

The organic growth option (also known as internal development) is a business growth method that would involve the expansion into a new geographic market (Bigland) using an internal project utilising our own competences and resources. We would be accountable to our existing shareholders (and not new ones which would be the case in an acquisition or joint venture).

Advantages of organic growth

Entering Bigland by ourselves would allow us to be in control and invest gradually, instead of having to make a large upfront investment (as would be the case for an acquisition). We would also avoid any issues relating to disagreements between ourselves and another company (as may arise with a joint venture). Organisational disruption would also be limited.

Disadvantages of organic growth

Customer acquisition in the new market might be slower as we would be unfamiliar to the electric vehicle battery producers in Bigland. Therefore, we may not be able to capitalise on this opportunity promptly. Depending on the market concentration of players, a slow response could be detrimental to our growth in the new market.

Also, we lack knowledge and experience of Bigland, which is a new geographic market for us. This would be a major operational risk and there would exist a high chance of failure. In contrast, this risk would be reduced if we entered the market via a joint venture, where the other party already has knowledge of the Bigland market.

There are financial risks too as there would be administration, marketing and sales costs. There would also be distribution costs of exporting our lithium product to the Bigland customers. That said, the financial risks would be lower than those associated with the acquisition or joint venture growth options.

Option 2 - Expansion via acquisition

Acquisition entails achieving growth by purchasing another company. In the context of an international expansion, this would involve the acquisition of a company that is already present in the new geographical area. Rotomyne could look into acquiring a lithium miner that successfully serves the types of customers that Rotomyne are targeting (i.e. electric vehicle battery producers in Bigland). Such a company would be familiar with the political and cultural environment of Bigland.

Advantages of acquisition growth

Acquisition is a quick way to enter into the international market and benefit from the increased demand for lithium. The lithium mining industry is highly competitive so acquisition of a foreign competitor reduces competition. This growth method would also allow us to achieve economies of scale more quickly (than organic growth).

Acquisition, while costly, may provide quicker access to local expertise and infrastructure. For instance, acquiring an established local mining company could help Rotomyne bypass the lengthy permitting processes and regulatory hurdles that come with setting up operations in a new country.

An acquisition would also enable Rotomyne access to a customer base and resources such as appropriately skilled staff (e.g. expertise and local knowledge), which may also be in short supply. An acquisition would provide value to Rotomyne through a combination of cost savings (e.g. shared overheads) and increased efficiency that we otherwise would not have if we expanded via organic growth.

Disadvantages of acquisition growth

However, an acquisition does have drawbacks. We could struggle to find a company that has the same strategic objectives and value (e.g. sustainability) as Rotomyne. We could also struggle to integrate the new company into ours due to the differences in organisational culture and systems.

An acquisition would require a significant upfront investment. In addition, it is expensive, both in terms of initial purchase price and the costs of post-merger integration. The latter may require high restructure and staff redundancy costs. There may also be hidden liabilities. Increased costs and paying too much for the target company would also erode any synergy benefits (such as combined marketing).

Also staff made redundant could breed contempt amongst the staff we keep, making integration more difficult. There may also be a clash in cultures (especially as the target company is 'foreign') that would further add to integration complexity. In addition, there may be systems integration issues which would cause operational disruption.

The international acquisition could become unviable due to legal and regulatory hurdles, such as anti-monopoly regulation, as well as exchange rate risk and increased debt risk from loans.

Bigland is noted as being politically unstable, so the political risk would also need to be assessed and controls put in place to mitigate it.

Option 3 - Expansion via joint venture

The option to grow via a joint venture requires a separate business entity being set up. Rotomyne and a joint venture partner would own shares in this entity. We would share control, assets, costs and profit based on our shareholding. The joint venture partner would ideally be a lithium miner that operates in Bigland.

Advantages of growth via a joint venture

A joint venture would involve less capital to expand into a foreign country like Bigland as compared to organic growth, and the risks of capital losses are shared.

We could potentially share our resources and access each other's knowledge or technology.

A local partner would allow us to focus on our strength, e.g. mining and refining lithium, while the local partner would take care of country-specific issues of regulation, culture, workforce, etc.

Disadvantages of growth via a joint venture

Due to the independence that company in a joint venture has, we do risk coming into conflict with our partner over strategic goals. These problems could be made worse depending on the controlling stake of the partner. A minority structure could leave us with little power, but a fifty-fifty structure could mean we enter a deadlock on strategic goals.

Lastly, there is our research skills which makes us innovative. We take pride in our research into battery improvements, which we have discovered through close collaboration with academics and customers. The conditions for entering a joint venture with a suitable company could involve us sharing some of our proprietary research. Therefore, Rotomyne would need to be mindful about how much we would be willing to give away to a joint venture partner in order for us to expand into Bigland.

3aii - Recommended expansion option

Out of the three options, it is recommended that a joint venture would be the most suitable growth option. This is because we could share the financial and operational risks with the joint venture partner. The partner company could also help navigate local regulations and reduce the political risk that is present in Bigland.

It is further recommended that the joint venture split (of control, assets, costs and profit) should be fifty-fifty. This is so that we would have an equal say in any decisions made. We do not want to be in a position that we lose control as then there will be a risk that our strategic objectives for the international expansion are not met, e.g. to maximise profitability.

Our global competitors are also using joint ventures to fulfil battery demand. For example, China-based Tianqi Lithium Corp. is partnering with CALB (China Aviation Lithium Battery). This joint venture focuses on battery-cell production and lithium salt refining to meet the increasing demand from Chinese electric vehicle battery manufacturers.

Acquisition is not recommended here as it would be more expensive than a joint venture. Also, it would be a challenge to find a suitable acquisition target that has the same strategic objectives and values as Rotomyne. We may also find that post merger integration of systems and culture would lead to delays, increased cost and this may ultimately erode any synergistic benefits.

Organic growth would also not be as suitable as a joint venture, as there would be a financial risk if the international expansion was to fail. The chance of failure would be high as we are not familiar with the new territory of Bigland as well as not being aware of the laws and regulation of the country. Bigland is also politically unstable, so we might not be able to manage the political risk alone.

Sub-task (b) - 45%: Business valuation

3bi - Business valuation methods evaluation

Asset-based valuation

Asset-based valuations depend on the sum of all the individual assets of the business. This might be represented by the net assets on the balance sheet (Statement of Financial Position), or an updated version of that, e.g. using current market values.

The asset-based valuation has little relevance in practice except in situations such as liquidations or disposal of certain parts of a business, as it ignores intangible assets (goodwill, human and intellectual capital) unless such capital appears in the company accounts. In effect, the net asset position does not reflect the future potential of the business.

Its suitability is limited when applied to Rotomyne as its main revenue is from lithium mining and refining. The future earnings from these do not appear in the statement of financial position. The finished goods stock of lithium could be valued based on its spot rate. However, the future value of the lithium deposits in the rock and brine mines would be ignored, so the asset-based valuation is inadequate.

Earnings-based valuation (P/E ratio method)

The earnings-based valuation using the P/E ratio method has the benefit that it places value on the future earnings of the company. In a stock-exchange listed entity, the P/E ratio is used to describe the relationship between the share price and earnings.

The P/E ratio can be calculated as follows: P/E ratio = market price per share divided by earnings per share. From this we can calculate the value of a business:

Value of a listed business = Earnings x P/E ratio

Note that the P/E ratio used in the valuation is often based on an industry specific P/E ratio that is an average of the P/E ratios of similar listed businesses. If the business being valued is not listed, the value of the company is reduced, typically by multiplying by 2/3; unlisted shares do not have an open market (i.e. they are illiquid) and it is likely that a lower value would have to be accepted if shares in the company were sold.

Rotomyne is a listed company and so such a devaluation factor would not need to be applied. Its valuation is calculated using an industry average P/E ratio (a benchmark P/E ratio). This however is not very accurate as this is an average of similar businesses whose P/E ratios may vary considerably.

A more appropriate P/E ratio would be that of a similar quoted entity (i.e. a proxy company), although that may be hard to find. This is because no two companies have the same product portfolios and brand identity. This uniqueness sets them apart and hence the P/E ratio of the proxy company would only be a guide.

Cash-flow based valuation

The future cash-flows of an entity discounted to the present value are a good indication of shareholder wealth. Thus, the forecasting of cash-flows and discounting at a risk adjusted discount rate is theoretically the most accurate valuation method. As such, this valuation method is most suitable for Rotomyne.

However, the valuation is based on the industry average cost of equity. This suffers from the same problems as noted above, namely that the average is composed of many values for cost of equity that may vary significantly. So the valuation is just a general guide.

In addition, included in the cost of equity is an element of return for financial risk in cases where companies have debt in their capital structure. The industry average will thus contain many entities with substantial debt finance. So the cost of equity is not specific to Rotmyne.

A more accurate cost of equity is required. As was highlighted in the P/E ratio method, a similar proxy company should be sought. Its cost of equity will be more relevant than that of the industry average. That said, the proxy company's cost of equity will have to be adjusted to reflect our capital structure.

It should also be noted that this approach is based on projected future earnings. Current estimates are for a 5% increase in perpetuity. We should ask on what basis those estimates are made and review their likely accuracy. Any change in the estimates used will have an impact on the cash-flows and hence the valuation.

We do know the estimates will be incorrect - the future will be different from estimates, but the question remains about whether they are an accurate estimate based on all of the information we currently have available. An independent evaluation may be useful here.

Recommended company valuation

The lowest valuation of Rotomyne is P$8,828m (asset based) and the highest valuation is P$22,584m (P/E based). The cash-flow based valuation of P$17,623m falls in between. The P/E based valuation is substantially higher than any of the other valuation methods.

Although the cash-flow method is theoretically correct, the reliability of the results is dependent on how accurately the cash-flows have been forecast as well as the discount factor used. It is also unrealistic to assume that the cash-flows will grow forever (i.e. in perpetuity).

As a result, none of the valuation figures are totally reliable. If a ballpark recommendation is required then the cash-flow based valuation should be used, i.e. a valuation of P$17,623m (and a share price of circa P$17.62). Our share price at the end of September 2024 is approximately $7, so it is undervalued based on the cash-flow based share price valuation. Any takeover bid that is below the cash-flow based valuation would have to be rejected.

Note that further detailed analysis would be required to get a more accurate valuation. For instance, the use of a proxy company's cost of equity that is then adjusted to match our capital structure. Also, we would need more accurate forecast cash-flows.

If you need anything else please let me know.

Kind regards,
SM

Marking Matrix

Task	Topic	Core activity	%	Marks	Total
1a	Risk register	D	55	18	
1b	Growth strategies	B	25	9	34
1c	Data security controls	A	20	7	
2a	PCA & Project control	E	55	18	33
2b	Currency risk types; Currency risk management	B	45	15	
3a	International expansion: Growth methods	A	55	18	33
3b	Business valuation methods	C	45	15	
					100

Marking guidance

Task 1		
(a)	D. Evaluate and mitigate risk	55%
(b)	B. Evaluate the business ecosystem and business environment	25%
(c)	A. Develop business strategy	20%

D. Evaluate and mitigate risk (18 marks)		55%
1ai - Describe the typical contents of a risk register, including how our principal risks log fits in.		12 marks
	No rewardable material	0
Level 1	Little knowledge of the risk register and risk log reuse shown and applied.	1-4
Level 2	Some knowledge of the risk register and risk log reuse shown and applied, and relates to the scenario.	5-8
Level 3	Good knowledge and understanding shown of the risk register and risk log reuse, relates to the scenario and references the pre-seen.	9-12

1aii - Discuss methods we could use to identify new risks.		6 marks
	No rewardable material	0
Level 1	Little knowledge of risk identification methods shown and applied	1-2
Level 2	Some knowledge of risk identification methods shown and applied, and relates to the scenario.	2-4
Level 3	Good knowledge and understanding shown of risk identification methods, relates to the scenario and references the pre-seen.	5-6

B. Evaluate the business ecosystem and business environment (9 marks)		25%
1bi - Discuss the different growth strategies that we might pursue.		9 marks
	No rewardable material	0
Level 1	Describes or states a few (1 to 2) strategies. No or little reference to risk level.	1-3
Level 2	Discusses all 4 strategies; States risk level (low, medium, high); Gives appropriate examples.	4-6
Level 3	Discusses all 4 strategies in full; Discusses risk level; Gives appropriate examples; References pre-seen.	7-9

A. Develop business strategy (7 marks)		20%
1ci - Discuss the controls that should be in place to improve data security.		**7 marks**
	No rewardable material	0
Level 1	States 1 to 2 controls. States some improvements.	1-3
Level 2	Explains controls. Explains improvement for each control; Relates to scenario.	4-5
Level 3	Explains controls; Provides example controls. Explains improvement for each control; Relates to scenario; References pre-seen.	6-7

Task 2		
(a)	E. Recommend and maintain a sound control environment	55%
(b)	B. Evaluate the business ecosystem and business environment	45%

E. Recommend and maintain a sound control environment (18 marks)		55%
2ai - Discuss the benefits of a PCA.		9 marks
	No rewardable material	0
Level 1	Describes benefits with no or little application.	1-3
Level 2	Discusses all benefits; Relates to scenario.	4-6
Level 3	Discusses all benefits in full; Relates to scenario; References pre-seen.	7-9
2aii - Recommend controls to manage a project's time, cost and quality.		9 marks
	No rewardable material	0
Level 1	Recommends a few (1 to 3) controls.	1-3

Level 2	Recommends several controls for each area; Justifies controls based on scenario.	4-6
Level 3	Recommends several controls for each area; Justifies controls based on scenario; Provides example controls.	7-9

B. Evaluate the business ecosystem and business environment (15 marks)		**45%**
2bi - Discuss the different types of currency risks.		**6 marks**
	No rewardable material	0
Level 1	Little knowledge of the currency risk types (economic, translation and transaction risk) shown and applied.	1-2
Level 2	Some knowledge of the currency risk types (economic, translation and transaction risk) shown and applied, and relates to the scenario.	3-4
Level 3	Good knowledge and understanding shown of the currency risk types (economic, translation and transaction risk), relates to the scenario and references the pre-seen.	5-6
2bii - Discuss methods to mitigate the foreign exchange transaction risk.		**9 marks**
	No rewardable material	0

Level 1	States 1 to 2 methods with basic critique.	1-3
Level 2	Discusses several methods (internal and external) with critique; Applies to scenario.	4-6
Level 3	Discusses several methods (internal and external) with critique; Applies to scenario; References pre-seen.	7-9

Task 3		
(a)	A. Develop business strategy	55%
(b)	C. Recommend financing strategies	45%

A. Develop business strategy (18 marks)		55%
3ai - Evaluate the different international expansion options.		12 marks
	No rewardable material	0
Level 1	Describes a few (1 to 2) options with basic critique.	1-4
Level 2	Evaluates several options; Relates to scenario.	4-8
Level 3	Evaluates all options in detail; Relates to scenario; References pre-seen.	9-12
3aii - Recommend, with reasons, the most appropriate expansion option for us.		6 marks
	No rewardable material	0

Level 1	Recommends an option. None or little justification.	1-2
Level 2	Recommends an option based on appropriateness to the scenario. Justifies in terms of why the option was chosen.	3-4
Level 3	Recommends an option based on appropriateness to scenario with more detail. Justifies in terms of why option chosen; States why other options not chosen; References pre-seen.	5-6

C. Recommend financing strategies (15 marks)		**45%**
3bi - Discuss the valuation methods used and their suitability, and recommend an approximate valuation.		**15 marks**
	No rewardable material	0
Level 1	Describes a few valuation methods. Recommends a method with none or little justification.	1-5
Level 2	Discusses several valuation methods and their suitability; Relates to scenario. Recommends an appropriate method with basic justification.	6-10
Level 3	Discusses all valuation methods and their suitability in detail; Relates to scenario; References pre-seen. Recommends an appropriate method with full justification.	11-15